Chapter Summary

Key Terms

5.1 Algebraic Expressions

- variable
- constant
- expression
- equation
- equal sign
- term
- coefficient
- like terms
- Distributive Property

5.2 Linear Equations in One Variable with Applications

- linear equation

5.3 Linear Inequalities in One Variable with Applications

- linear inequality
- Addition and Subtraction Property of Linear Inequalities
- Multiplication and Division Property of Linear Inequalities

5.4 Ratios and Proportions

- ratio
- proportion
- constant of proportionality
- scale
- construct ratios
- solve proportions
- use proportions to solve scaling problems

5.5 Graphing Linear Equations and Inequalities

- ordered pair
- origin
- points on the axes
- linear equation in two variables
- standards form of a linear equation
- solution
- linear inequality in two variables
- solution to a linear inequality
- boundary line

5.6 Quadratic Equations In One Variable with Applications

- monomial
- polynomial
- binomial
- trinomial
- quadratic equation
- Zero Product Property

5.7 Functions

- relation
- domain
- function
- mapping
- vertical line test

5.8 Graphing Functions

- intercepts of a line
- slope
- slope-intercept form

5.9 Systems of Linear Equations in Two Variables

- system of linear equations
- solutions of a system of equations
- contradictions
- identities
- coincident lines
- consistent system of linear equations
- inconsistent system of linear equations

5.10 Systems of Linear Inequalities in Two Variables

- system of linear inequalities

5.11 Linear Programming

- linear programming
- objective function
- constraint

Key Concepts

5.1 Algebraic Expressions

- Algebra is useful because it allows us to understand many situations in real life by modeling them with expressions.
- Algebraic expressions are the building blocks of algebra. From algebraic expressions we can create algebraic equations.
- Algebraic expressions are often simplified and evaluated using the four arithmetic operations.

5.2 Linear Equations in One Variable with Applications

- Solving linear equations means discovering what the value of the variable in a linear equation represents in the given conditions.
- When solving a linear equation, most often you will have one solution; however, a linear equation may have no solutions or infinitely many solutions.

5.3 Linear Inequalities in One Variable with Applications

- Inequalities can be used when the possible values (answers) in a certain situation are numerous, or when the exact value (answer) is not known, but it is known to be within a range of possible values.
- Linear inequalities can be represented using a number line or using interval notation.

5.4 Ratios and Proportions

- A ratio is a comparison of two numbers. The ratio of two numbers a and b can be written as: a to b OR $a{:}b$ OR the fraction a/b.
- All fractions are ratios, but not all ratios are fractions. Ratios make part to part, part to whole, and whole to part comparisons. Fractions make part to whole comparisons only.
- When two ratios are equal, we say they are in proportion or are proportional.
- Setting up proportions allows us to solve many various situations where three of the four values of the proportion are known.

5.5 Graphing Linear Equations and Inequalities

- Linear equations can be represented graphically on a rectangular coordinate system.
- Solving linear equations in two variables means finding the point where two lines intersect. There are three possibilities: The lines intersect at exactly one point; the lines do not intersect (they are parallel); or the lines intersect everywhere (they are the same line).
- Solving linear inequalities in two variables means finding a region of possible answers. Every point in this region will make both inequalities true statements.

- Plotting points is a standard way to help graph linear equations and linear inequalities.

5.6 Quadratic Equations In One Variable with Applications

- A quadratic equation is an algebraic equation where the highest power (degree) of the equation is two.
- To solve a quadratic equation is to find the value(s) that when substituted in for the variables, will make the equation equal to zero.
- There can be two, one, or no solutions to any quadratic equation.
- There are several methods to solve a quadratic equation. These methods include factoring quadratic equations, graphic quadratic equations, using the square root method, and using the quadratic formula.

5.7 Functions

- A relation is any set of ordered pairs (x, y). All of the x-values of the set are the domain, and all of the y-values of the set are the range.
- A relation is a function if each x-value in the domain is assigned to exactly one element in the range. A y-value in the range can have more than one x-value assigned to it; but each x-value can only be assigned to one y-value.
- For the function $y = f(x)$, f is the name of the function, x is the domain value variable, and $y = f(x)$ is the range value variable.
- The vertical line test is a test that can be done on the graph of a relation to determine if it is a function.

5.8 Graphing Functions

- Every linear function can be graphically represented by a unique line that shows all the solutions of the equation.
- The points where the graph of a line intersects the x-axis and y-axis are called the intercepts of the line.
- Most lines will have one x-intercept and one y-intercept. Only if the line is straight vertical (no y-intercept) or straight horizontal (no x-intercept) will it not have both intercepts. Note that a line that is straight vertical is not a function, but a line that is straight horizontal is a function.
- Since any two points determine a straight line, any linear function can be graphed if both intercepts are known.
- The slope of a linear function is the ratio of the vertical change divided by the horizontal change. It is often referred to as $\frac{rise}{run}$.
- A formula for finding the slope of linear functions is $\frac{y_2 - y_1}{x_2 - x_1}$, for any two points of the linear function (x_1, y_1) and (x_2, y_2).

5.9 Systems of Linear Equations in Two Variables

- To solve a system of linear equations means finding the point or points where the two linear equations intersect.
- Two lines can intersect at one point, no points if they are parallel, or every point if they are the same equation.
- Systems of linear equations can be solved by graphing, by using substitution, or by using the elimination method.

5.10 Systems of Linear Inequalities in Two Variables

- To solve a system of linear inequalities means to find the area(s) where the points in that area make all the linear inequalities true.
- Systems of linear inequalities can be solved by graphing the linear equations associated with the inequalities, then 'testing' points to see whether the values of the point make the equation true or not.

5.11 Linear Programming

- Linear programming is a mathematical technique to solve problems involving finding maximums or minimums where a linear function is limited by various constraints.
- An objective function is a linear function in two or more variables that describes the quantity that needs to be maximized or minimized.
- In linear programming, a constraint is a restriction that affects the maximum or minimum values of an objective function.
- Through the creation of objective functions and restraints, a linear system can be developed and solved through linear programming.

Videos

5.1 Algebraic Expressions

- Q&A: Why We Teach Algebra (https://openstax.org/r/Teach_Algebra)

5.2 Linear Equations in One Variable with Applications

- Solving for a Variable in an Equation (https://openstax.org/r/Solving_for_a_variable)

5.5 Graphing Linear Equations and Inequalities

- Graphing Linear Inequalities in Two Variables (https://openstax.org/r/Graphing_linear)

5.6 Quadratic Equations In One Variable with Applications

- Factoring with the Box Method (Area Model) (https://openstax.org/r/Factoring_with_the_Box)
- Solving Quadratics with the Zero Property (https://openstax.org/r/Zero_Property)
- Solving Quadratics with the Quadratic Formula (https://openstax.org/r/Solving_Quadratics)

5.7 Functions

- Relations and Functions (https://openstax.org/r/Relationsp_and_Functions)
- Domain and Range on Graphs (https://openstax.org/r/Domain)

5.9 Systems of Linear Equations in Two Variables

- Practice with Solving Applications of Systems of Equations (https://openstax.org/r/Practice_with_Solving)
- Applications of Systems of Linear Equations (https://openstax.org/r/Applications_of_Systems)

5.10 Systems of Linear Inequalities in Two Variables

- Solving Systems of Linear Inequalities by Graphing (https://openstax.org/r/Solving_Systems)
- Systems of Linear Inequalities (https://openstax.org/r/Systems_of_Linear)

Formula Review

5.1 Algebraic Expressions

- Distributive Property: $a(b+c) = ab + ac$

5.3 Linear Inequalities in One Variable with Applications

- For any numbers a, b, and , if $a < b$, then $a + c < b + c$ and $a - c < b - c$.
- For any numbers a, b, and c, if $a > b$, then $a + c > b + c$ and $a - c > b - c$.
- For any numbers a, b, and c,
  multiply or divide by a positive:
  if $a < b$ and $c > 0$, then $ac < bc$ and $\frac{a}{c} < \frac{b}{c}$
  if $a > b$ and $c > 0$, then $ac > bc$ and $\frac{a}{c} > \frac{b}{c}$
  multiply or divide by a negative:
  if $a < b$ and $c < 0$, then $ac > bc$ and $\frac{a}{c} > \frac{b}{c}$
  if $a > b$ and $c < 0$, then $ac < bc$ and $\frac{a}{c} < \frac{b}{c}$

5.8 Graphing Functions

- To calculate slope (m), use the formula
  $m = \frac{rise}{run}$,
  where the rise measures the vertical change and the run measures the horizontal change.
- To find the slope of the line between two points (x_1, y_1) and (x_2, y_2), use the formula
  $m = \frac{y2-y1}{x2-x1}$

Projects

Ratio and Proportion—Comparing Prices, Part 1

Go to your favorite coffee shops and find out what a same sized drink costs at each. You can do something similar for pizza as well. Find the unit rate (i.e., price per ounce or price per square inch). For example, go to your favorite coffee place and find the price per units on all their large coffee drinks. Or go to your favorite pizza place and compare prices of all their extra-large pizzas (by price per square inch). Write a report on the best deals.

Ratio and Proportion—Comparing Prices, Part 2

Rather than comparing prices of different, but same sized drinks (or pizzas), compare unit prices of the same drinks but of different sizes. Find out what the best bargain is based on price per ounce, price per square inch, etc. For example, compare the prices of your favorite soft drink sold at a local store, but in various sizes (i.e., 12-ounce can, 16-ounce bottle, 20-ounce bottle, 1-liter bottle, and multipacks). Or go to a pizza place and find out what the best bargain is on their menu, based on price per square inch of pizza. Write a report on the best deals.

Systems of Linear Inequalities—Comparing Cell Phone Plans

Go to the websites of different cell phone companies and compare their plans. Write a report on “the best deals. "Best Deals” doesn’t necessarily mean “cheapest.” You will need to look at what each company provides concerning restrictions (constraints) on minutes to talk. What are the constraints on the cell phone coverage for each company? Do they cover your area of the country well? Do they cover the entire United States well, or at least areas where you will be travelling? Is this coverage 5G, or is it less? Can you add a phone easily? Can you bring your previous phone number to this plan? The possibilities of constraints affecting each plan are several. So your task is to determine which plan is best, based on not only cost but also all constraints you deem important.

Chapter Review

Algebraic Expressions

1. Translate from algebra to words: $2x + 3$
2. Translate from words to algebra: the quotient of $5x$ and 7.
3. Translate from an English phrase to an expression: A gym charges $5.00 per class c and a $20 membership fee.
4. Use parentheses to make the following statement true: $2^2 \div +5 - 4 \times 3 = -5$
5. Evaluate and simplify $x^2 - 5x$ when $x = 5$.
6. Perform the indicated operation for the expression: $(3x^2 + 2x + 1) + (x^2 - 2x + 2 - (x^2 - 3x + 11)$

Linear Equations in One Variable with Applications

7. Solve the linear equations using a general strategy: $3x - 7 = 8$
8. Solve the linear equations using properties of equations: $(5x - 3) + 7 = 9x - 2(x + 7)$
9. It costs 30 cents for an ear of corn. Construct a linear equation and solve how much it costs to buy 23 ears of corn.
10. State whether the following equation has exactly one solution, no solution, or infinitely many solutions $5x + 4 = 7x - 14$
11. Solve the formula $A = \frac{(b_1 + b_2)}{2}h$ for b_1

Linear Inequalities in One Variable with Applications

12. Graph the inequality $x > 8$ on a number line and write the interval notation.
13. Solve the inequality $-2x < -6$, graph the solution on the number line, and write the solution in interval notation.
14. Construct a linear inequality to solve the application: Daniel wants to surprise his girlfriend with a birthday party at her favorite restaurant. It will cost $42.75 per person for dinner, including tip and tax. His budget for the party is $500. What is the maximum number of people Daniel can have at the party?

Ratio and Proportions

Christer opened a bag of marbles and counted the number of each color. They found they had 9 green, 4 yellow, 13 black, 11 orange, 8 blue, and 7 red.

15. What is the ratio of green marbles to orange marbles?
16. What is the ratio of red marbles to blue marbles?
17. What is the ratio of sum of the marbles with an odd number of marbles to the sum of the marbles with an even number of marbles?
18. What is the ratio of black marbles to all marbles?

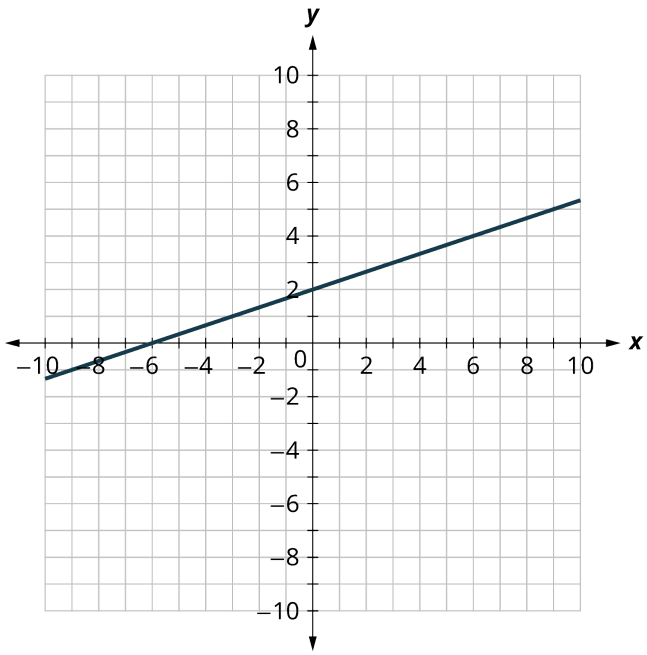
19. Solve: $\frac{7}{21} = \frac{21}{x}$
20. Basil the cat is 17 pounds and 24 inches long from head to tail. In his new movie *Claws*, he is supersized to 50 pounds. How many inches long will he be? Round your answer to the nearest tenth.

Graphing Linear Equations and Inequalities

21. For each ordered pair, decide:
    I. Is the ordered pair a solution to the equation?
    II. Is the point on the line in the graph shown?

    $y = \frac{1}{3}x + 2$

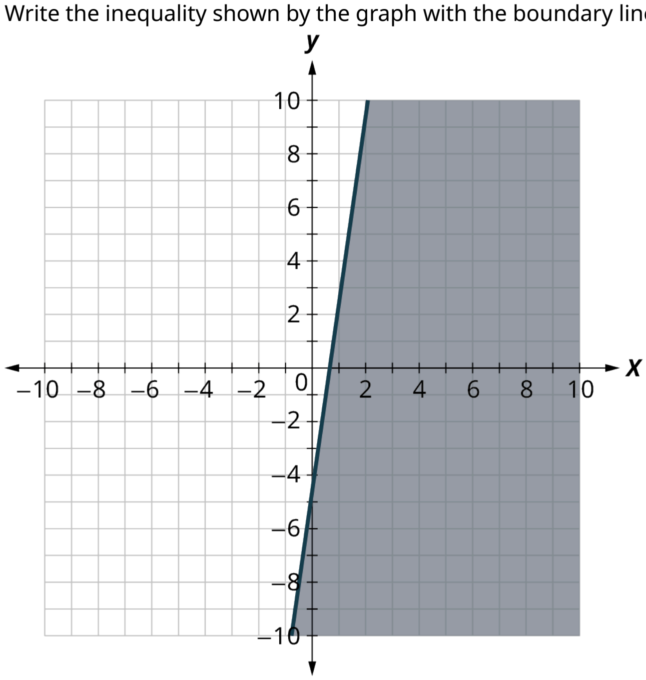
    A: $(0, -3)$ B: $(-3, 1)$; C: $(-2, -4)$; D: $(0, 2)$



22. Graph $y = \frac{1}{2}x + 5$ by plotting points.

23. Determine whether each ordered pair is a solution to the inequality $y > 2x + 3$.
A: $(0, 0)$ B: $(2, 1)$ C: $(-1, -5)$ D: $(-6, -3)$ E: $(1, 0)$

24. Write the inequality shown by the graph with the boundary line $y = 7x - 5$.



25. Graph the linear inequality: $y < \frac{1}{3}x + 7$.

Quadratic Equations In One Variable with Applications

26. Multiply $(3x - 1)(x + 7)$.

27. Factor $x^2 + 8x - 9$.

28. Graph and list the solutions to the quadratic equation $x^2 + 11x + 18 = 0$.

29. Solve $x^2 + 7x + 12 = 15x - 3$ by factoring.

30. Solve $169x^2 = 25$ using the square root method.

31. Solve $x^2 + 7x + 3 = 0$ using the quadratic formula.

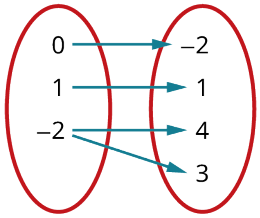
32. The height in feet, h, of an object shot upwards into the air with initial velocity, v_0, after t seconds is given by the formula $h = -16t^2 + v_0 t$. A firework is shot upwards with initial velocity 130 feet per second. How many seconds will it take to reach a height of 260 feet? Round to the nearest tenth of a second.

Functions

33. Evaluate the function $f(x) = x^2 - 3x + 21$ at the values $f(-2)$, $f(-1)$, $f(0)$, (1), and $f(2)$.

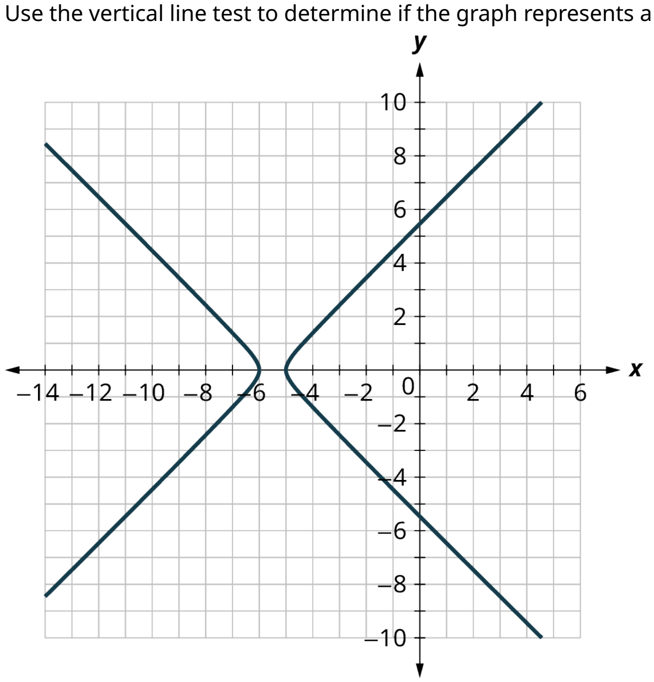
34. Determine whether $\{(-2, 31), (-1, 25), (1, 19), (2, 19)\}$ represents a function.

35. Determine whether the mappings represent a function:



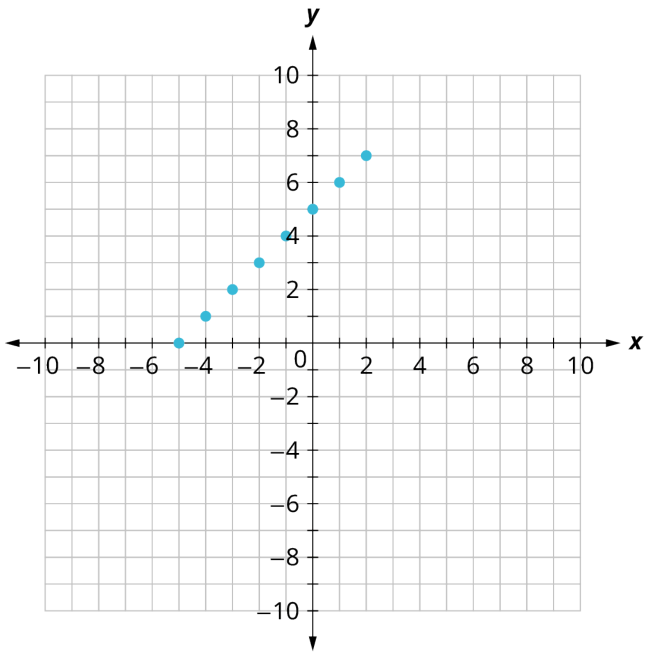
36. Determine whether $5x^2 + 2y^2 = 10$ represent y as a function of x.

37. Use the vertical line test to determine if the graph represents a function.



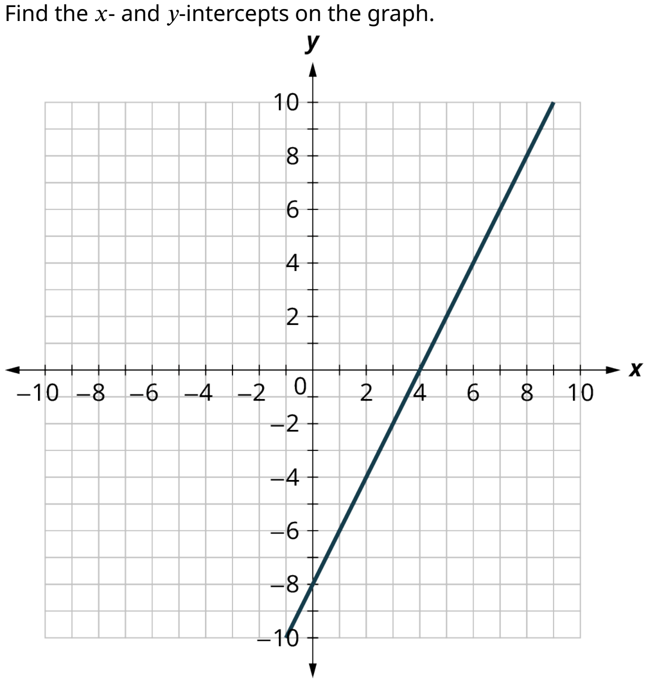
38. Use the set of ordered pairs to find the domain and the range.
$\{(-3, 9), (-2, 4), (-1, 1), (0, 0), (1, 1), (2, 4), (3, 9)\}$

39. Use the graph shown to find the domain and the range.



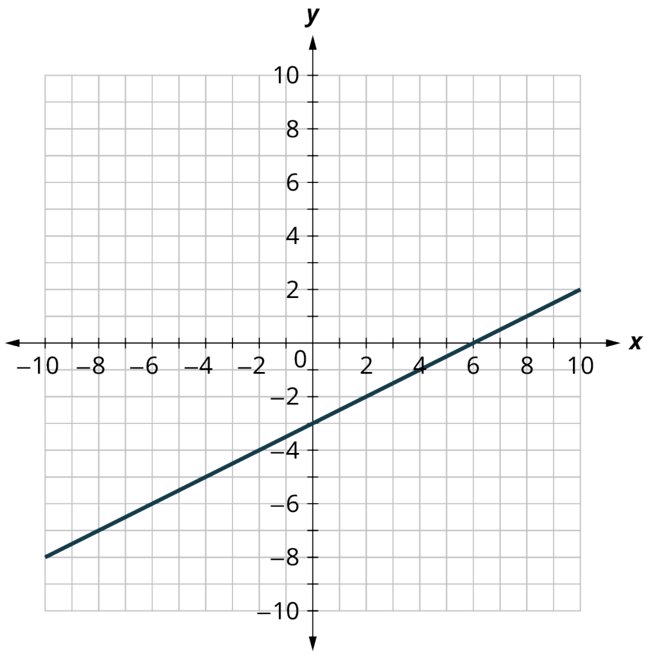
Graphing Functions

40. Find the x- and y-intercepts on the graph.



41. Graph $y = 3x - 3$ using the intercepts.

42. Find the slope of the line in the graph shown.



43. Use the slope formula to find the slope of the line between (2, 4) and (5, 7).

44. Identify the slope and y-intercept of $2y = -\frac{2}{3}x + 10$.

45. Graph the line of $2y = -\frac{2}{3}x - 4$ using its slope and y-intercept.

The equation $P = 28 + 2.54w$ models the relation between the monthly water bill payment, P, in dollars, and the number of units of water, w, used.

46. Find the payment for a month when 0 units of water were used.
47. Find the payment for a month when 15 units of water were used.
48. Interpret the slope and P-intercept of the equation.
49. Graph the equation.

System of Linear Equations in Two Variables

50. Determine if (0, 1) and (2, 3) are solutions to the given system of equations.

$$\begin{cases} x - 2y = 0 \\ 3x - y = 5 \end{cases}$$

51. Solve the system of equations by graphing.

$$\begin{cases} y = 3x + 1 \\ 6x - y = 2 \end{cases}$$

52. Solve the system of equations by substitution.

$$\begin{cases} y = 7x + 2 \\ 3x - y = -6 \end{cases}$$

53. Solve the systems of equations by elimination.

$$\begin{cases} -5x + y = -15 \\ 3x + y = 17 \end{cases}$$

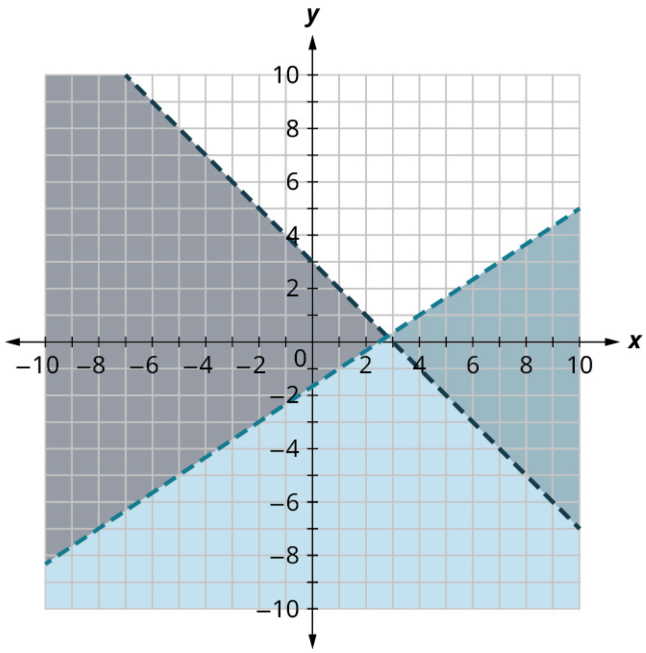
54. Kenneth currently sells suits for company A at a salary of \$22,000 plus a \$10 commission for each suit sold. Company B offers him a position with a salary of \$28,000 plus a \$4 commission for each suit sold. How many suits would Kenneth need to sell for the options to be equal?

Systems of Linear Inequalities in Two Variables

55. Determine whether (0, 0) and (2, 3) are solutions to the system.

$$\begin{cases} 3x + y > 5 \\ 2x - y \leq 10 \end{cases}$$

56. Determine whether (0, 0) and (6, –8) are solutions to the darkest shaded region of the graph.



57. Solve the systems of linear equations by graphing.

$$\begin{cases} y < 3x + 1 \\ 6x - y \geq 2 \end{cases}$$

Jocelyn desires to increase both her protein consumption and caloric intake. She desires to have at least 35 more grams of protein each day and no more than an additional 200 calories daily. An ounce of cheddar cheese has 7 grams of protein and 110 calories. An ounce of parmesan cheese has 11 grams of protein and 22 calories.

58. Write a system of inequalities to model this situation.

59. Graph the system.

60. Could she eat 1 ounce of cheddar cheese and 3 ounces of parmesan cheese?

61. Could she eat 2 ounces of cheddar cheese and 1 ounce of parmesan cheese?

Linear Programming

A toy maker makes two plastic toys, the Ring (x) and the Stick (y). The toy maker makes $4 per Ring and $6 per Stick. The Ring uses 3 feet of plastic, while the Stick uses 5 feet of plastic. Today the toy maker has 40 feet of plastic available. The toy maker also only makes 10 plastic toys per day. To maximize profit, how many of each toy should the toy maker make?

62. Find the objective function.

63. Write the constraints as a system of inequalities.

64. Graph of the system of inequalities.

65. Find the value of the objective function at each corner point of the graphed region.

66. To maximize his profit, how many of each toy should the toymaker make?

Chapter Test

1. Perform the indicated operation for the expression: $(3x^2 + 2x + 1) - (x^2 - 2 + 2) + x(3x + 11)$

2. Solve the linear equations using properties of equations: $x(x - 2) + 5 = 9x + x(x - 6)$

3. It costs 55 cents for a stamp. Construct a linear equation and solve how much it cost to buy 50 stamps.

4. Solve the formula $A = \frac{(b_1 + b_2)}{2}h$ for h.

5. Solve the inequality $-2x < -6$, graph the solution on the number line, and write the solution in interval notation.

6. Construct a linear inequality to solve the application: Bella wants to buy a round of shakes for her friends. It will cost $4.75 per shake, including tip and tax. Her budget is $50. What is the maximum number of friends Bella can buy shakes for?

7. *Manneken Pis* is a famous statue in Brussels, Belgium. It is 24 inches tall and weighs 37.5 pounds. The average man is 69 inches tall and weighs 198 pounds. Is *Manneken Pis* proportional to the average male?

8. Graph $y = \frac{1}{4}x + 1$ by plotting points.

9. Graph the linear inequality: $y \geq 2x + 7$

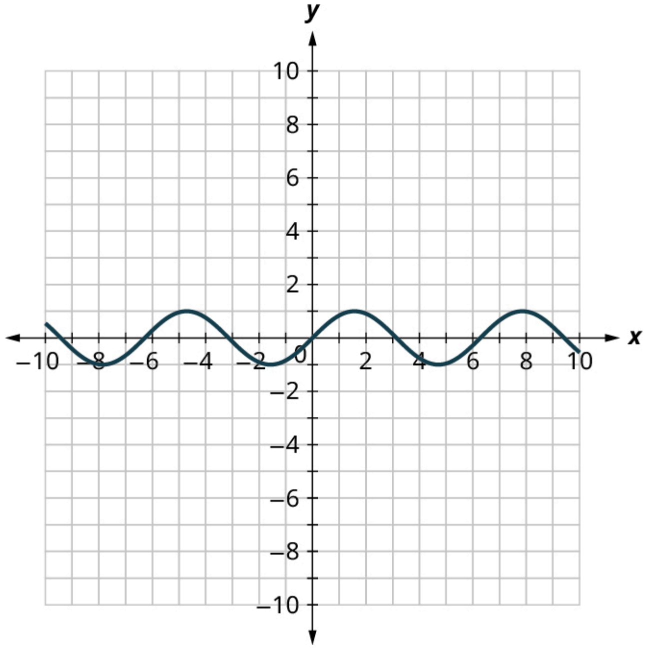
10. Graph and list the solutions to the quadratic equation $x^2 - 36 = 0$.

11. Solve $x^2 - 49 = 0$ by factoring.

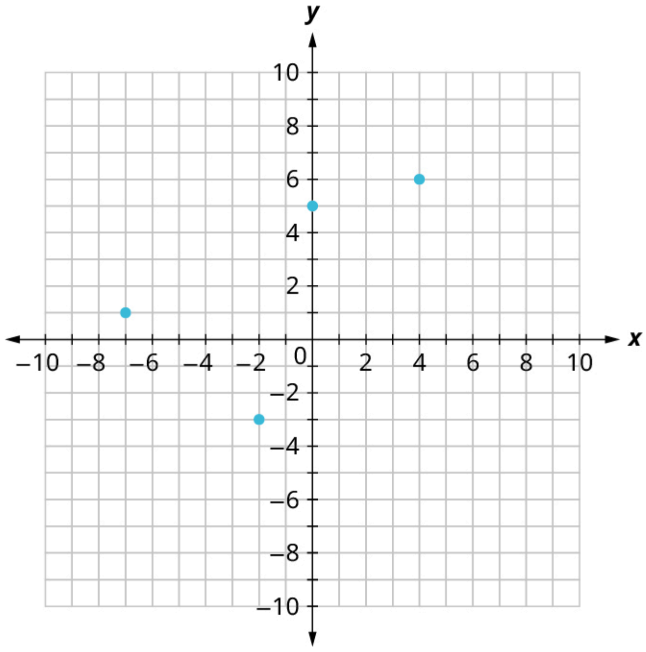
12. Solve $x^2 + 5x + 2 = 0$ using the quadratic formula.

13. Evaluate the function $f(x) = -3x + 21$ at the values $f(-2)$, $f(-1)$, $f(0)$, $f(1)$, and $f(2)$.

14. Use the vertical line test to determine if the given graph represents a function.



15. Use the graph shown to find the domain and the range.



16. Graph $2y = x - 4$ using the intercepts.

17. Use the slope formula to find the slope of the line between (1, 4) and (3, 5).

18. Identify the slope and y-intercept of $y - \frac{2}{3}x = 1$.

19. Graph the line of $y = \frac{2}{3}x + 1$ using its slope and y-intercept.

The equation $C = 4.50 + 15p$, models the cost of visiting the Cat Café in San Diego for one hour. C, in dollars, is the total cost and the cost per person, p, is $15 plus a $4.50 reservation fee.

20. Find the payment for two people.
21. Find the payment for five people.
22. Interpret the slope and C-intercept of the equation.
23. Graph the equation.

24. Solve the system of equations by graphing.
$$\begin{cases} y - 3x = 3 \\ y = -6x + 12 \end{cases}$$

25. Solve the system of equations by substitution.
$$\begin{cases} y = 5x - 5 \\ -3x + y = -3 \end{cases}$$

26. Solve the systems of equations by elimination.
$$\begin{cases} y = \frac{1}{3}x - 6 \\ y = x - 3 \end{cases}$$

27. Anna goes to the concession stand at a movie theater. She buys 5 popcorns and 4 large sodas and pays a total of $60. During intermission, Isabelle goes to the concession stand. She buys 1 popcorn and 2 large sodas and pays a total of $18. What is the cost of one popcorn, and the cost of one large soda?

28. Solve the systems of linear equations by graphing.
$$\begin{cases} y > \frac{1}{3}x - 1 \\ y < x - 3 \end{cases}$$

Juliette is selling fresh lemonade and cupcakes. She sells a cup of lemonade for $2 and a cupcake for $3. She needs to make at least $100 to donate to the local cat sanctuary. She needs to sell at least 20 cups of lemonade.

29. Write a system of inequalities to model this situation.
30. Graph the system.
31. Could she sell 30 cups of lemonade and 10 cupcakes and make $100?
32. Could she sell 20 cups of lemonade and 30 cupcakes and make $100?

A toy maker makes exactly two toys out of wood; the Box (x) and the Bat (y). He makes $5 per Box and $6 per Bat. Each Box requires 30 ounces of wood, and each Bat requires 45 ounces of wood. Today the toy maker has 270 ounces of wood available. The toy maker also only makes 8 wooden toys per day. To maximize profit, how many of each wooden toy should the toy maker make?

33. Find the objective function.
34. Write the constraints as a system of inequalities.
35. Graph of the system of inequalities.
36. Find the value of the objective function at each corner point of the graphed region.
37. To maximize profit, how many of each toy should the toymaker make?